
ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH

Department of Information Technology

Synopsis

On

API Code Integration for NAMASTE Portal & ICD-11 TM2

1. Introduction

1. Overview

The healthcare ecosystem in India consists of both traditional AYUSH systems (Ayurveda, Siddha, Unani) and modern biomedicine. However, these systems currently operate with different diagnostic coding standards, creating a gap in interoperability and data exchange. The NAMASTE (National AYUSH Morbidity and Standardized Terminologies Electronic) system provides standardized terminology for traditional medicine, while the WHO's ICD-11 (including Traditional Medicine Module 2 – TM2) offers globally recognized disease classification.

This project focuses on designing a FHIR-compliant API-based microservice that integrates NAMASTE with ICD-11 (TM2 and Biomedicine) into Electronic Medical Record (EMR) systems. The system enables dual coding of diagnoses, allowing clinicians to record both traditional and biomedical perspectives in a unified digital format. This integration enhances clinical documentation, interoperability, and compliance with India's EHR standards.

2. Purpose

The main purpose of this project is to develop a standardized and interoperable API solution that bridges the gap between AYUSH and modern medical systems. The objectives are as follows:

- **Interoperability:** Enable seamless integration of NAMASTE and ICD-11 codes within EMR systems using FHIR standards.

- **Dual Coding Support:** Allow clinicians to record both traditional and biomedical diagnoses in a single patient record.
- **Standard Compliance:** Ensure adherence to Indian EHR Standards (FHIR R4, OAuth 2.0, SNOMED CT, LOINC).
- **Improved Clinical Decision Support:** Provide accurate and standardized terminology mapping for better diagnosis and treatment.
- **Healthcare Analytics & Claims:** Support insurance claims and government analytics using globally accepted ICD-11 coding.

2. Literature Survey

1. Existing Problem

The current healthcare systems, especially in India's AYUSH sector, face several critical challenges due to the lack of standardized and interoperable terminology integration:

- **Lack of Interoperability:** Traditional medicine systems (Ayurveda, Siddha, Unani) use NAMASTE codes, while modern healthcare relies on ICD-11. These systems are not properly integrated, creating a communication gap.
- **Limited EMR Support:** Most Electronic Medical Record (EMR) systems are designed mainly for biomedical data and do not support AYUSH-based terminology effectively.
- **Absence of Dual Coding:** There is no efficient mechanism to record both traditional and biomedical diagnoses together in a single patient record.
- **Manual and Fragmented Processes:** Many AYUSH healthcare providers still rely on paper-based or semi-digital systems, leading to inefficiencies and data inconsistency.
- **Compliance Challenges:** Meeting Indian EHR standards (FHIR R4, OAuth, SNOMED CT, LOINC) becomes difficult without proper integration frameworks.

2.2 Proposed Solution

To overcome these limitations, this project proposes a **FHIR-based API microservice** that integrates NAMASTE with ICD-11 (TM2 and Biomedicine) into EMR systems.

- **Terminology Integration:** Develop a system that ingests NAMASTE data and synchronizes it with WHO ICD-11 APIs for unified access.
- **Dual Coding Mechanism:** Enable automatic mapping between NAMASTE and ICD-11 TM2 codes, allowing clinicians to record both codes simultaneously.
- **RESTful API Services:** Provide endpoints for auto-complete search, code lookup, and translation between NAMASTE and ICD-11.
- **FHIR Compliance:** Store and exchange healthcare data using FHIR resources such as CodeSystem, ConceptMap, and Problem List.
- **Secure Access:** Implement OAuth 2.0 authentication using ABHA tokens along with audit trails for security and compliance.
- **User Interface / CLI Tool:** Offer a simple interface for clinicians to search terms, view mappings, and create standardized medical records.

This solution ensures interoperability, improves clinical efficiency, and aligns AYUSH systems with global healthcare standards

3. Theoretical Analysis

1. Block Diagram

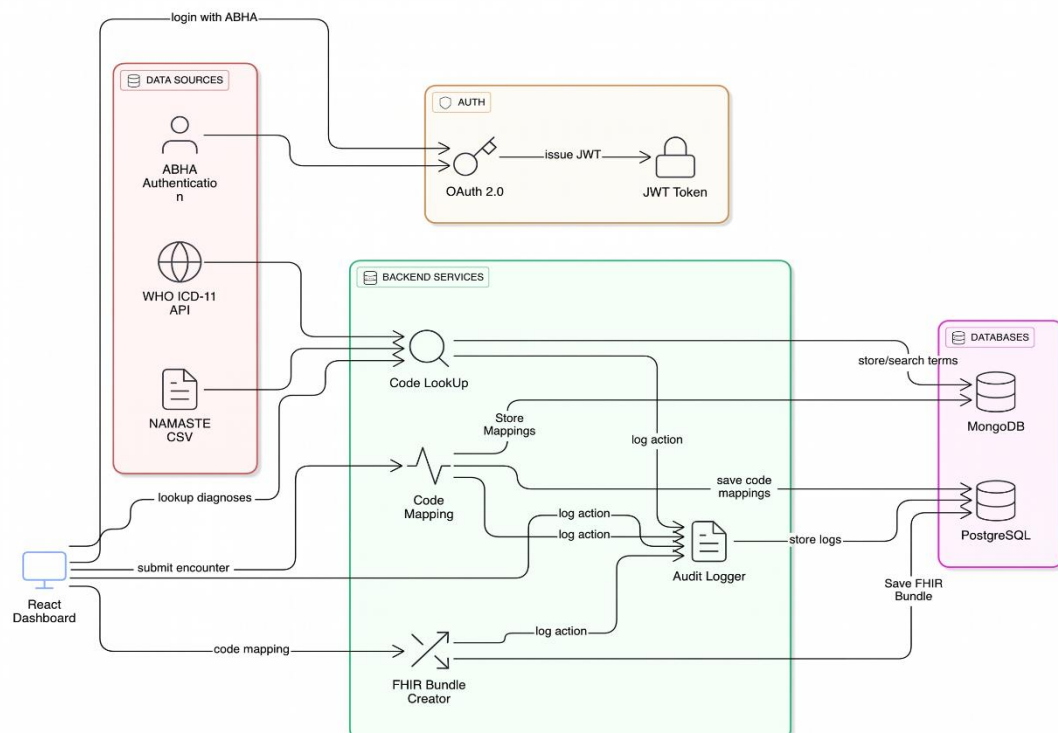


Fig 1.1 Technical Architecture

3.2 Hardware/Software Designing

Hardware Requirements:

- System with minimum **8 GB RAM** (16 GB recommended)
- Processor: **Intel i5 / i7 or equivalent**
- Stable internet connection for API communication
- Storage for datasets (NAMASTE + ICD-11 data)

Software Requirements:

- **Frontend:**
 - HTML, CSS, JavaScript / React (optional)
- **Backend:**
 - Node.js / Python (FastAPI / Flask)
- **FHIR Tools:**
 - HAPI FHIR Server or any FHIR R4 compliant framework
- **Database:**
 - PostgreSQL / MongoDB
- **APIs & Integration:**
 - WHO ICD-11 API
 - NAMASTE CSV dataset
- **Security:**
 - OAuth 2.0 (ABHA integration)
- **Other Tools:**
 - Postman (API testing)
 - Git/GitHub (version control)

4. Applications

The proposed system has multiple real-world applications in the healthcare and digital health ecosystem:

- **AYUSH Healthcare Systems:**

Enables doctors practicing Ayurveda, Siddha, and Unani to digitally record diagnoses using standardized NAMASTE codes along with ICD-11 mapping.

- **Hospitals & EMR Platforms:**
Can be integrated into existing EMR/EHR systems to support dual coding and improve clinical documentation.
- **Insurance & Claims Processing:**
Helps in generating globally accepted ICD-11 codes, making AYUSH treatments eligible for insurance claims.
- **Government Health Programs:**
Assists organizations like the Ministry of AYUSH in collecting accurate morbidity data for policy-making and analytics.
- **Research & Data Analytics:**
Provides standardized datasets for medical research, epidemiology studies, and AI-based healthcare innovations.

REFERENCES

- [NAMASTE Portal](#)
- [WHO ICD-11 API documentation](#)
- [WHO ICD-11 Reference Guide \(Chapter 26 Traditional Medicine Module 2\)](#)
- [FHIR Terminology Service specification](#)
- [FHIR ConceptMap documentation](#)
- [ABHA \(Ayushman Bharat Health Account\) API integrator guide](#)
- [India's 2016 EHR Standards \(Ministry of Health & Family Welfare\)](#)
- [Building healthcare microservices: a FHIR-native approach](#)

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